

EfficientNet-B0 Region-Graph Attention Network with a Fuzzy Confidence Head for Five-Class Cervical Cytology Classification

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Abstract—Distinguishing cervical-cell categories is challenging because classes with different clinical meanings may still look alike in terms of texture, staining, nuclear shape, and cytoplasmic appearance. Ensemble approaches can improve stability by combining several convolutional networks, but they also raise the computational burden and commonly merge only the final class probabilities, after much of the spatial detail has already been compressed. This study introduces an EfficientNet-B0 Region-Graph Attention Network with a Fuzzy Confidence Head for classifying isolated cervical cells into five categories. A single ImageNet-pretrained EfficientNet-B0 encoder produces a $1280 \times 7 \times 7$ feature map, which is projected to 160 channels and represented as a five-node graph formed by four spatial quadrants and one global-context node. Two graph attention layers allow local and global descriptors to exchange information before graph mean pooling and multilayer-perceptron classification. The optimization objective also incorporates a differentiable fuzzy confidence-and-separation score. Evaluation is carried out on all 4,049 SIPaKMeD images using stratified five-fold cross-validation. The completed runs achieve mean accuracy, macro precision, macro recall, and macro F1-score of 97.46%, 97.48%, 97.47%, and 97.47%, respectively. Across the combined out-of-fold predictions, 3,946 samples are classified correctly and 103 are misclassified. In a descriptive comparison with the assigned fuzzy-rank ensemble, which reports 95.43% five-class accuracy, the proposed method gains 2.03 percentage points in accuracy and lowers the error rate by approximately 44.3%. The paper includes quantitative results, qualitative explanations, confidence and error analyses, and an ablation discussion that remains faithful to the executed code. Overall, the findings suggest that explicit local-global feature interaction can deliver strong performance with a single compact encoder, while external validation, group-aware data splitting, calibration, and controlled retraining of individual components are still required before any clinical application.

The implementation, configuration files, fold-level results, and reproducibility materials are available at: <https://github.com/nayab20im-tech/EfficientNet-GAT-Fuzzy-Cytology>.

Index Terms—cervical cytology, Pap smear, EfficientNet-B0, graph attention network, fuzzy confidence, SIPaKMeD, Grad-CAM, medical image classification

I. INTRODUCTION

Cervical cancer continues to be a major public-health concern, although timely screening and treatment of precancerous changes can prevent a large proportion of cases [1]. Automated cytology analysis may assist specialists by drawing attention to suspicious cells and reducing the workload associated with repeated manual review. For such a system to be useful,

however, it must separate visually similar categories while also remaining efficient and reasonably interpretable.

Pap-smear classification is a fine-grained visual-recognition task. Dyskeratotic and Koilocytotic cells belong to abnormal categories, Metaplastic cells are benign, and Parabasal together with Superficial-Intermediate cells are normal. In practice, these classes are not always visually distinct. Differences in staining, orientation, contrast, background content, and the relative appearance of the nucleus and cytoplasm can make separate categories resemble one another. A model that reduces the whole image to a single global representation may therefore lose relationships between diagnostically useful regions.

The assigned baseline proposed by Manna *et al.* [3] fuses the outputs of Inception-v3, Xception, and DenseNet-169 with fixed fuzzy-rank functions. Although the ensemble performs well, running three separate encoders is computationally expensive. Its fusion also takes place only after each network has generated class probabilities, so evidence from different image regions cannot influence one another during representation learning. This limitation motivates a single-backbone design that keeps coarse spatial information and explicitly models the connection between local regions and the overall cell appearance.

To address these issues, this work develops an EfficientNet-B0-GAT-Fuzzy framework. EfficientNet-B0 serves as the visual backbone, four quadrant descriptors together with one global descriptor define a small graph, graph attention learns image-specific weights between connected nodes, and a fuzzy confidence-and-separation term is included during optimization. The description remains aligned with the supplied notebook and does not add components that were not implemented.

The main contributions are:

- A cervical-cell classifier built around one EfficientNet-B0 backbone and a lightweight 1×1 projection, replacing the three-CNN baseline ensemble.
- A compact five-node graph that retains coarse spatial information through four local quadrants and one global-context node.
- Two masked graph attention layers that learn how local and global evidence should be combined for each image before classification.

- A fuzzy confidence-and-separation head and training objective that match the code, supported by five-fold out-of-fold evaluation, Grad-CAM visualization, confidence and error studies, and an ablation discussion that reports only executed or analytically verifiable findings.

II. LITERATURE REVIEW

A. Cervical Cytology Datasets and CNN Classification

Plissiti *et al.* [2] introduced SIPaKMeD to support both feature-based and image-based recognition of normal and pathological cervical cells. The dataset contains 4,049 isolated-cell images distributed across five cytological classes and is now widely used in automated Pap-smear research. Earlier deep-learning work established that convolutional networks can learn useful cervical-cell representations directly from image data. Lin *et al.* [4] also demonstrated that combining morphology with appearance can improve discrimination, highlighting the diagnostic value of both nuclear and cytoplasmic structure.

Researchers have also examined transfer learning and explicit feature-selection strategies. Basak *et al.* [5] combined deep representations with principal-component analysis and Grey Wolf Optimization, obtaining strong results on several cervical datasets. A drawback of this type of pipeline is that representation learning, feature selection, and classification remain separate processing stages rather than being optimized as one end-to-end model.

B. Ensemble and Fuzzy Decision Methods

Manna *et al.* [3] combined Inception-v3, Xception, and DenseNet-169 through a fuzzy rank-based ensemble and reported 95.43% accuracy on the five-class SIPaKMeD task; this method is used as the assigned baseline. Pramanik *et al.* [6] later introduced a fuzzy distance-based ensemble for cervical cancer detection. Together, these studies indicate that fuzzy fusion can improve robustness. At the same time, using several complete backbones increases both training and inference cost, and the fusion is typically applied to final predictions rather than to spatially organized feature representations.

C. Attention, Transformers, and Structured Reasoning

EfficientNet jointly scales depth, width, and image resolution through a compound strategy [7]. Its B0 variant is a practical choice when a strong visual encoder is needed without a very large parameter count. Attention-based cervical classification has also gained interest. CerviFormer, for example, applies cross-attention and a latent Transformer to integrate Pap-smear information [9]. More recent geometry-aware attention research likewise suggests that structural priors and long-range dependencies can be valuable for cervical cytology [10].

Graph neural networks offer another way to model structure by treating image regions as nodes and the relations between them as edges. Graph Attention Networks (GATs) use masked attention to assign different weights to neighboring nodes [8]. This idea is relevant to cytology because a diagnosis may

depend on how the nucleus, cytoplasm, boundary, and overall cell appearance relate to one another, rather than on a single isolated area. The proposed method therefore uses a small fixed region graph instead of a dense pixel-level graph, keeping the computation manageable while still enabling relational reasoning.

D. Explainability and Reliability

A high accuracy value does not explain which parts of an image influenced the decision. Grad-CAM traces gradients into a convolutional layer to create a coarse map of visually important regions [11]. Here, it is applied to held-out samples to examine whether the full CNN–GAT–fuzzy pipeline focuses on meaningful cellular structures or on potential shortcuts such as image borders and background patterns. Confidence distributions and confidently misclassified cases are also studied, since a wrong prediction made with high certainty presents a greater reliability concern than an uncertain error.

III. PROPOSED METHODOLOGY

A. System Overview

For an RGB Pap-smear image $\mathbf{x} \in \mathbb{R}^{3 \times 224 \times 224}$, the implemented processing sequence is

$$\mathbf{x} \rightarrow E_\theta \rightarrow \mathbf{F}_0 \rightarrow \mathbf{F} \rightarrow \mathbf{H} \rightarrow \text{GAT}_{1,2} \rightarrow \mathbf{z} \rightarrow \mathbf{o} \rightarrow \mathbf{s} \rightarrow \hat{y}. \quad (1)$$

These stages represent image preprocessing, EfficientNet-B0 feature extraction, channel reduction, region-graph formation, graph-attention updates, graph pooling, MLP classification, and fuzzy decision scoring.

B. Preprocessing and EfficientNet-B0 Encoder

All images are resized to 224×224 . Training samples undergo random horizontal flipping, rotation within $\pm 12^\circ$, and light colour jitter. The images are then converted to tensors and normalized using ImageNet channel statistics, whereas validation images follow the same resizing and normalization steps without random augmentation.

The classification layer of EfficientNet-B0 is discarded, leaving only the convolutional feature extractor. Its output is

$$\mathbf{F}_0 = E_\theta(\mathbf{x}) \in \mathbb{R}^{1280 \times 7 \times 7}. \quad (2)$$

A learnable 1×1 convolution then reduces the channel count from 1280 to 160:

$$\mathbf{F} = \text{Conv}_{1 \times 1}(\mathbf{F}_0) \in \mathbb{R}^{160 \times 7 \times 7}. \quad (3)$$

This projection lowers the dimensionality supplied to the graph module without changing the 7×7 spatial arrangement.

C. Five-Node Region Graph

The projected map is split at its integer midpoints into top-left, top-right, bottom-left, and bottom-right regions. Because the map size is 7×7 , these regions measure 3×3 , 3×4 , 4×3 , and 4×4 , respectively. Adaptive average pooling turns each region into a 160-dimensional descriptor, while a fifth descriptor is produced by pooling the entire feature map:

$$\mathbf{h}_r = \text{GAP}(\mathbf{F}_{\Omega_r}), \quad r \in \{\text{TL}, \text{TR}, \text{BL}, \text{BR}, \text{G}\}. \quad (4)$$

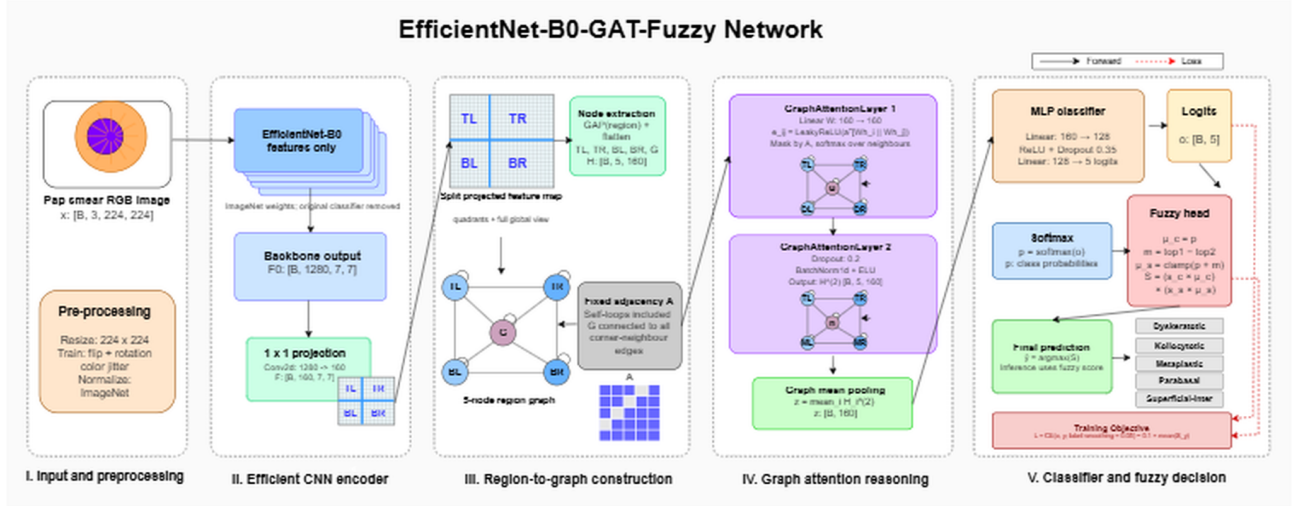


Fig. 1. Overview of the proposed EfficientNet-B0-GAT-Fuzzy pipeline. The 224×224 RGB cell image passes through EfficientNet-B0, a 160-channel projection, four local quadrant nodes and one global node, two graph attention layers, graph pooling, an MLP classifier, and the softmax and fuzzy confidence outputs.

The node matrix is

$$\mathbf{H} = [\mathbf{h}_{TL}, \mathbf{h}_{TR}, \mathbf{h}_{BL}, \mathbf{h}_{BR}, \mathbf{h}_G]^T \in \mathbb{R}^{5 \times 160}. \quad (5)$$

The notebook defines the graph with the following fixed adjacency matrix:

$$\mathbf{A} = \begin{bmatrix} 1 & 1 & 1 & 0 & 1 \\ 1 & 1 & 0 & 1 & 1 \\ 1 & 0 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 \end{bmatrix}. \quad (6)$$

Each node has a self-loop. A local node is linked to its horizontal and vertical neighbors as well as to the global node, but diagonally opposite local regions are not directly connected.

D. Graph Attention Reasoning

Within a graph attention layer, node i is first mapped through a linear transformation

$$\mathbf{u}_i = \mathbf{W}\mathbf{h}_i. \quad (7)$$

For every permitted edge (i, j) , the layer computes the following unnormalized attention score:

$$e_{ij} = \text{LeakyReLU}(\mathbf{a}^\top [\mathbf{u}_i \parallel \mathbf{u}_j]), \quad (8)$$

where $\parallel$ indicates concatenation. Scores associated with absent edges are masked with $-\infty$, after which attention is normalized across the neighbors of node i :

$$\alpha_{ij} = \frac{\exp(e_{ij})}{\sum_{k: A_{ik}=1} \exp(e_{ik})}. \quad (9)$$

The resulting representation for node i is

$$\mathbf{h}'_i = \text{Dropout} \left[\text{ELU} \left(\text{BN} \left(\sum_{j: A_{ij}=1} \alpha_{ij} \mathbf{u}_j \right) \right) \right]. \quad (10)$$

The model stacks two single-head $160 \rightarrow 160$ GAT layers and then averages the resulting representations across all five nodes:

$$\mathbf{z} = \frac{1}{5} \sum_{i=1}^5 \mathbf{h}_i^{(2)} \in \mathbb{R}^{160}. \quad (11)$$

E. MLP Classifier and Fuzzy Confidence Head

Classification is performed with a $160 \rightarrow 128 \rightarrow 5$ MLP applied to the pooled graph vector:

$$\mathbf{o} = \mathbf{W}_2 \text{Dropout}_{0.35} [\text{ReLU}(\mathbf{W}_1 \mathbf{z} + \mathbf{b}_1)] + \mathbf{b}_2. \quad (12)$$

The corresponding softmax probabilities are

$$\mathbf{p} = \text{softmax}(\mathbf{o}). \quad (13)$$

Let $p_{(1)}$ and $p_{(2)}$ denote the largest and second-largest probabilities, and define

$$m = p_{(1)} - p_{(2)} \quad (14)$$

as the separation margin between them. For class k , the implementation uses the memberships

$$\mu_{c,k} = p_k, \quad (15)$$

$$\mu_{s,k} = \text{clip}(p_k + m, 0, 1). \quad (16)$$

Two trainable global scalars, γ_c and γ_s , are used to obtain

$$s_k = (\gamma_c \mu_{c,k})(\gamma_s \mu_{s,k}), \quad (17)$$

with the final class selected as

$$\hat{y} = \arg \max_k s_k. \quad (18)$$

In total, the network has 4,286,243 trainable parameters. Of these, 4,007,548 belong to the EfficientNet feature extractor, while the remaining 278,695 are distributed across the projection layer, two GAT layers, MLP, and fuzzy scalars.

Algorithm 1 Five-fold model training and out-of-fold assessment

Require: Dataset $\mathcal{D}$, labels $\mathbf{y}$, folds $K = 5$, maximum epochs $E = 14$
Ensure: Selected checkpoints, fold-level metrics, out-of-fold predictions, and qualitative explanations

- 1: Generate stratified folds using shuffling and random seed 42
- 2: **for** $f = 1$ to K **do**
- 3: Divide $\mathcal{D}$ into $\mathcal{D}_{tr}^f$ for training and $\mathcal{D}_{va}^f$ for validation
- 4: Initialize the EfficientNet-B0–GAT–Fuzzy model with ImageNet-pretrained backbone weights
- 5: Configure AdamW with backbone LR 8×10^{-5} and head LR 4×10^{-4}
- 6: **for** $e = 1$ to E **do**
- 7: **for** each mini-batch $(\mathbf{x}, y)$ in $\mathcal{D}_{tr}^f$ **do**
- 8: Obtain logits, softmax probabilities, and fuzzy scores
- 9: Compute $\mathcal{L}$ using Eq. (19)
- 10: Update the model parameters with mixed-precision AdamW
- 11: **end for**
- 12: Measure validation accuracy and macro-averaged metrics
- 13: Step ReduceLROnPlateau and retain the best-performing checkpoint
- 14: **if** validation accuracy shows no improvement for four epochs **then**
- 15: **break**
- 16: **end if**
- 17: **end for**
- 18: Restore the selected checkpoint
- 19: Generate one prediction for each sample in $\mathcal{D}_{va}^f$
- 20: **end for**
- 21: Combine the held-out predictions and produce the quantitative and qualitative analyses

TABLE I
SIPAKMED CLASS DISTRIBUTION.

Class	Images
Dyskeratotic (abnormal)	813
Koilocytotic (abnormal)	825
Metaplastic (benign)	793
Parabasal (normal)	787
Superficial-Intermediate (normal)	831
Total	4,049

F. Training Objective

The training loss combines label-smoothed cross-entropy with a reward term based on the fuzzy score of the ground-truth class:

$$\mathcal{L} = \text{CE}_{LS}(\mathbf{o}, y; \epsilon = 0.05) - \lambda_f s_y, \quad \lambda_f = 0.10. \quad (19)$$

Since the reward term is subtracted, the overall objective can fall below zero during later epochs. Such values do not indicate an invalid probability or numerical failure; they follow directly from the form of the implemented loss.

IV. EXPERIMENTAL SETUP

A. Dataset

Experiments are conducted on the five-class SIPaKMeD isolated-cell dataset [2]. The class counts obtained by the notebook are presented in Table I.

B. Cross-Validation Protocol

The evaluation follows stratified five-fold cross-validation with shuffling and random seed 42. Folds 1–4 use 3,239 images for training and 810 for validation, whereas fold 5 uses 3,240 training images and 809 validation images. Each image is assigned exactly one out-of-fold prediction from a model for which that image was never part of the training set. The resulting predictions are then used for pooled error

TABLE II
EXECUTED TRAINING CONFIGURATION.

Setting	Value
Input size	224×224
Cross-validation	5-fold stratified
Random seed	42
Batch size	16
Maximum epochs	14
Backbone learning rate	8×10^{-5}
New-layer learning rate	4×10^{-4}
Optimizer	AdamW
Weight decay	10^{-4}
Scheduler	ReduceLROnPlateau
Scheduler factor / patience	0.5 / 2 epochs
Early-stopping patience	4 epochs
Label smoothing	0.05
GAT dropout	0.20
MLP dropout	0.35
Fuzzy reward coefficient	0.10
Mixed precision	Enabled with CUDA

analysis, class-wise evaluation, confidence analysis, and Grad-CAM visualization.

Because the available folders do not include patient or slide identifiers, the splits are created at image level. The reported evaluation should therefore not be treated as patient-level or slide-level external validation.

C. Implementation and Training Configuration

The notebook was run on Kaggle with Python 3.12.13, PyTorch 2.10.0+cu128, and CUDA support. The code relies on torchvision, scikit-learn, NumPy, pandas, Pillow, and Matplotlib. Automatic mixed precision is activated whenever a CUDA device is available [13].

AdamW separates weight decay from the gradient-based update rule [12]. Using different learning rates lets the pretrained backbone adapt gradually while the newly initialized layers learn more quickly.

D. Evaluation Metrics

Performance is reported using accuracy, macro precision, macro recall, and macro F1-score. If C is the confusion matrix and $K = 5$ is the number of classes, then

$$\text{Accuracy} = \frac{\sum_k C_{kk}}{\sum_i \sum_j C_{ij}}, \quad (20)$$

$$\text{Precision}_{\text{macro}} = \frac{1}{K} \sum_{k=1}^K \frac{TP_k}{TP_k + FP_k}, \quad (21)$$

$$\text{Recall}_{\text{macro}} = \frac{1}{K} \sum_{k=1}^K \frac{TP_k}{TP_k + FN_k}, \quad (22)$$

$$F1_{\text{macro}} = \frac{1}{K} \sum_{k=1}^K \frac{2P_k R_k}{P_k + R_k}. \quad (23)$$

Macro averaging gives the same weight to each class, preventing the larger categories from dominating the overall metric.

V. IMPLEMENTATION AND REPRODUCIBILITY

A. Layer-Wise Tensor and Parameter Profile

Table III presents the tensor dimensions and the exact distribution of trainable parameters across the implemented model. EfficientNet-B0 contributes 4,007,548 parameters, or roughly 93.5% of the full network. The 1×1 projection adds 204,960 parameters and is the largest newly initialized block because it maps all 1,280 backbone channels into the 160-dimensional graph representation. Each GAT layer contains 26,240 parameters, and the MLP contains 21,253. Region partitioning, node assembly, adjacency masking, and graph mean pooling introduce no trainable parameters, while the fuzzy head adds only two scalars. Together, these values match the total of 4,286,243 trainable parameters recorded in the notebook.

This parameter breakdown helps explain why the proposed architecture is lighter than a three-backbone ensemble. Only one full image encoder is retained, and the relational module contributes about 1.2% of the total parameter count. Parameter count alone, however, is not a complete measure of speed because runtime also depends on graph preparation, data transfer, and low-level kernel execution. The notebook does not contain standardized measurements of latency, throughput, or energy consumption, so this paper does not introduce estimated inference values. A valid efficiency comparison would require every model to be tested on the same hardware, with the same batch size, numerical precision, and warmed-up evaluation procedure.

B. Checkpoint Selection and Out-of-Fold Reconstruction

A separate model is initialized for each fold using ImageNet-pretrained EfficientNet-B0 weights. The backbone and the newly added components are placed in different AdamW parameter groups, allowing the pretrained encoder to use the smaller learning rate. At the end of every epoch, performance is measured on the held-out fold, the scheduler is updated, and the checkpoint with the highest validation accuracy is kept. Training stops after four consecutive epochs without improvement or when the 14-epoch limit is reached. The selected checkpoints occur at epochs 5, 8, 11, 8, and 6 for folds 1–5, respectively.

For the final aggregate and qualitative analyses, each fold checkpoint is reloaded and applied only to the images excluded from its training set. The five held-out prediction files are then merged so that every image in the dataset has one out-of-fold record. This avoids explaining or evaluating a sample with a checkpoint that was trained on that same sample. It also makes it possible to trace class-wise scores, confidence distributions, difficult correct cases, confident errors, and Grad-CAM examples to the specific held-out model that produced them.

C. Saved Outputs and Auditability

The workflow saves training histories, fold metrics, confusion matrices, classification reports, selected checkpoints, and a combined out-of-fold prediction table. All figures in

the report are generated from these saved outputs rather than being entered manually. Keeping model execution separate from report generation allows the plots to be reproduced without rerunning training and enables each displayed sample to be linked to its true label, predicted label, fold, softmax probabilities, and fuzzy score.

The implementation uses random seed 42 and creates shuffled stratified folds. The saved environment information lists Python 3.12.13, PyTorch 2.10.0 with CUDA 12.8 support, together with torchvision, scikit-learn, NumPy, pandas, Pillow, and Matplotlib. Mixed precision is used only when CUDA is present. Reproducibility could be strengthened further by releasing the exact fold indices, locked dependency versions, hardware details, deterministic-backend settings, checkpoint hashes, and any available patient or slide identifiers. Repeating the experiment with several initialization seeds would also help distinguish variation caused by fold difficulty from variation caused by optimization.

D. Scope of Reproducible Claims

All numerical claims are limited to results produced by the supplied implementation and the analyses derived from its out-of-fold predictions. The full five-fold experiment was completed, but alternative structures—including one GAT layer, no GAT layers, removal of the global node, and training without the fuzzy reward—were not retrained. They are therefore listed only as planned controlled ablations, with no invented performance values. The published baseline is also used only for descriptive comparison because it was not rerun on the same saved folds. Maintaining this distinction separates measured evidence from mathematical observations and from experiments that have yet to be performed.

VI. RESULTS AND DISCUSSION

A. Quantitative Five-Fold Results

The fold-level measurements are reported in Table IV. Accuracy varies between 95.93% and 98.52%, producing a mean of 97.46% and a population standard deviation of 0.89 percentage points. Fold 2 gives the highest result, whereas fold 1 is the most difficult.

The learning curves indicate that strong validation performance is usually reached within relatively few epochs. Differences between folds show that the composition of the held-out subset affects task difficulty, which supports using cross-validation instead of relying on a single train–test partition.

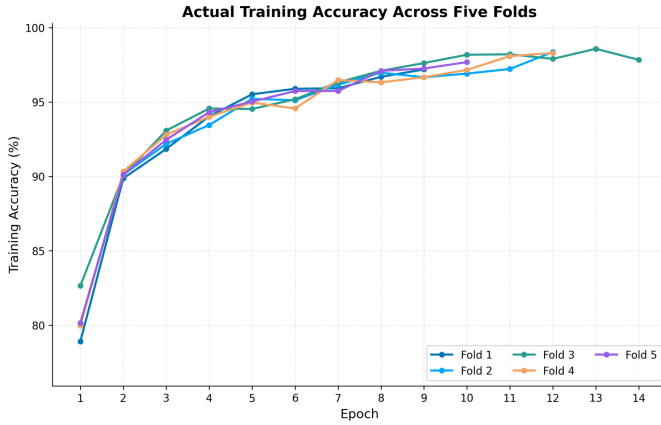
Across all folds, the training objective declines steadily, whereas the validation objective fluctuates more as the model approaches convergence. This is reasonable because each held-out partition contains a different combination of visually ambiguous examples. Checkpoints are chosen by validation accuracy rather than by the minimum value of the modified loss, so the fuzzy reward alone cannot determine model selection. The stopping points also confirm that the folds require different amounts of optimization, with the best checkpoints appearing between epochs 5 and 11. The reported average

TABLE III
IMPLEMENTED TENSOR FLOW AND EXACT TRAINABLE-PARAMETER ALLOCATION.

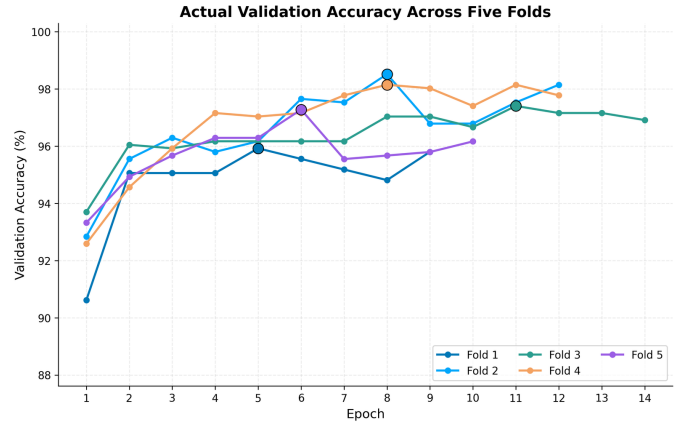
Module	Output representation	Parameters	Function
EfficientNet-B0 feature extractor	$1280 \times 7 \times 7$	4,007,548	Extracts convolutional features that are adapted to the normalized RGB cervical-cell image.
1×1 projection	$160 \times 7 \times 7$	204,960	Compresses the channel dimension without discarding the final spatial arrangement.
Quadrant and global pooling	5×160	0	Forms the TL, TR, BL, BR, and global nodes through adaptive average pooling.
Graph attention layer 1	5×160	26,240	Carries out the first masked exchange of image-specific local and global information.
Graph attention layer 2	5×160	26,240	Further updates the node features after the initial relational reasoning step.
Graph mean pooling	160	0	Combines the five updated nodes into a single graph-level descriptor.
MLP classifier	5 logits	21,253	Maps the graph descriptor to five logits through the nonlinear $160 \rightarrow 128 \rightarrow 5$ classifier.
Fuzzy confidence head	5 scores	2	Applies the two trainable global scales for confidence and class separation.
Complete model	5 class scores	4,286,243	Complete end-to-end EfficientNet-B0–GAT–Fuzzy model.

TABLE IV
FIVE-FOLD RESULTS. PRECISION, RECALL, AND F1 ARE MACRO AVERAGED.

Fold	Best epoch	Train	Validation	Accuracy (%)	Precision (%)	Recall (%)	F1 (%)
1	5	3,239	810	95.93	95.97	95.94	95.94
2	8	3,239	810	98.52	98.52	98.53	98.53
3	11	3,239	810	97.41	97.44	97.41	97.42
4	8	3,239	810	98.15	98.17	98.16	98.16
5	6	3,240	809	97.28	97.31	97.30	97.29
Mean	7.60	3,239.20	809.80	97.46	97.48	97.47	97.47
Std. dev.	2.06	0.40	0.40	0.89	0.88	0.89	0.89



(a) Training accuracy.



(b) Validation accuracy.

Fig. 2. Training and validation accuracy across the five folds. Curves end at different epochs because of early stopping.

therefore comes from independently selected models rather than from repeatedly evaluating a single checkpoint.

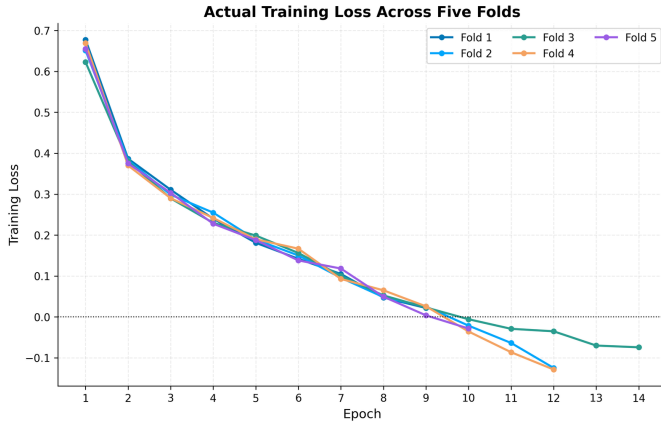
B. Aggregate Out-of-Fold Evaluation

Combining the five held-out partitions yields exactly one prediction for each of the 4,049 images. The pooled performance is summarized in Table V.

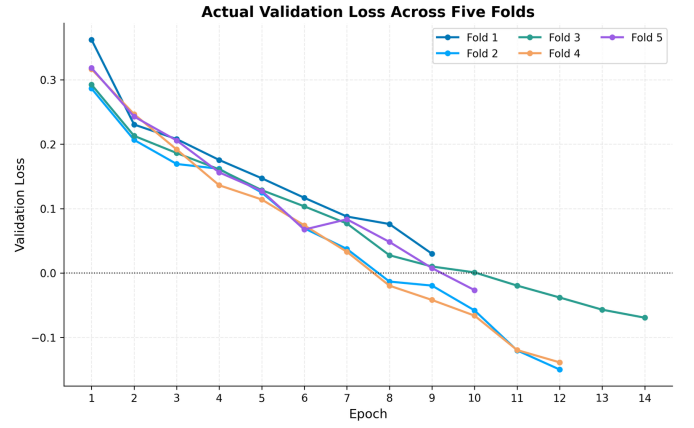
C. Comparison with the Assigned Baseline

For five-class SIPaKMeD classification, the assigned fuzzy-rank ensemble reports 95.43% accuracy, 95.34% precision, 95.38% recall, and 95.36% F1-score [3]. Table VI places these published values beside the mean obtained from the completed proposed-model runs.

The baseline error rate is 4.57%, while the proposed mean error rate is 2.54%. Therefore, the relative reduction in error

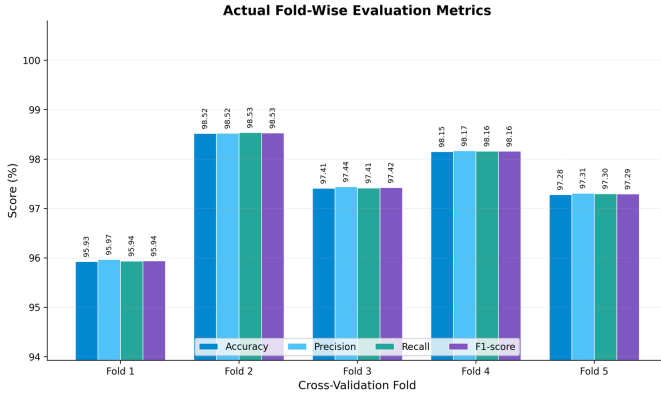


(a) Training objective.

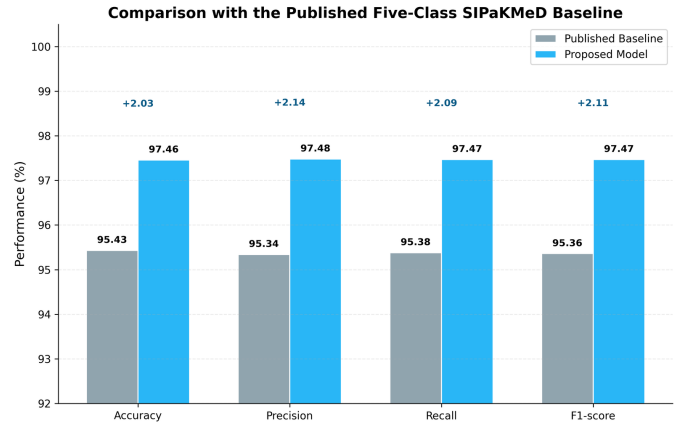


(b) Validation objective.

Fig. 3. Objective values recorded for all five folds. Because the loss subtracts the ground-truth fuzzy reward from label-smoothed cross-entropy, later values can become negative without implying numerical instability.



(a) Fold-wise metrics.



(b) Baseline versus proposed mean.

Fig. 4. Summary of the quantitative results. The baseline comparison remains descriptive because the baseline was not evaluated on the same stored folds.

TABLE V
AGGREGATE OUT-OF-FOLD EVALUATION.

Measure	Result
Total images	4,049
Correct predictions	3,946
Incorrect predictions	103
Accuracy	97.46%
Macro precision	97.47%
Macro recall	97.47%
Macro F1-score	97.47%

TABLE VI
DESCRIPTIVE COMPARISON WITH THE ASSIGNED BASELINE.

Method	Accuracy	Precision (%)	Recall (%)	F1 (%)
Fuzzy-rank ensemble [3]	95.43	95.34	95.38	95.36
Proposed model	97.46	97.48	97.47	97.47
Absolute gain	+2.03	+2.14	+2.09	+2.11

TABLE VII
AGGREGATE OUT-OF-FOLD CLASS-WISE METRICS.

Class	Precision (%)	Recall (%)	F1 (%)	Support
Dyskeratotic	97.05	97.05	97.05	813
Koilocytotic	95.22	94.18	94.70	825
Metaplastic	95.91	97.60	96.75	793
Parabasal	100.00	99.11	99.55	787
Superficial-Intermediate	99.16	99.40	99.28	831

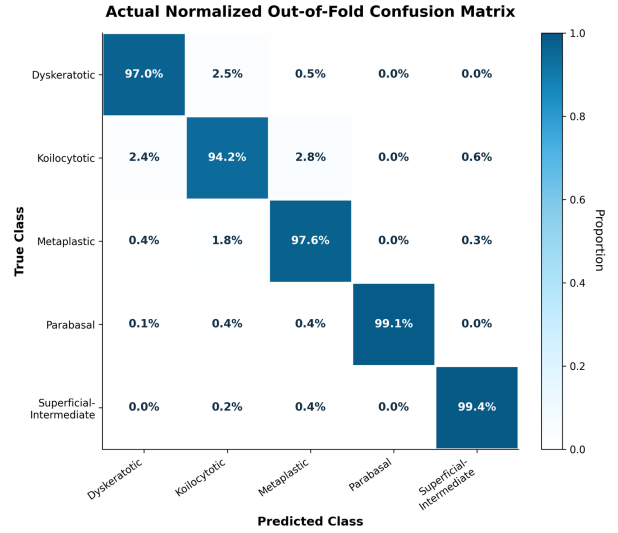
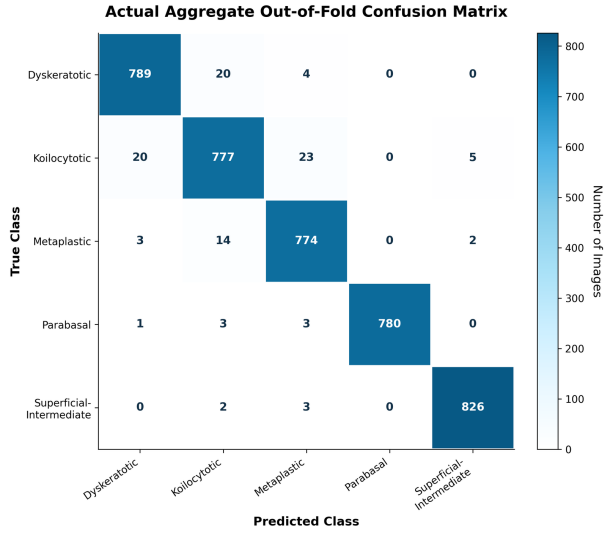
The difference is substantial as a descriptive result, but it should not be interpreted as a controlled statistical comparison. The baseline numbers come from the published study, and the two methods were not evaluated using identical fold assignments in the supplied notebook.

D. Class-Wise Quantitative Analysis

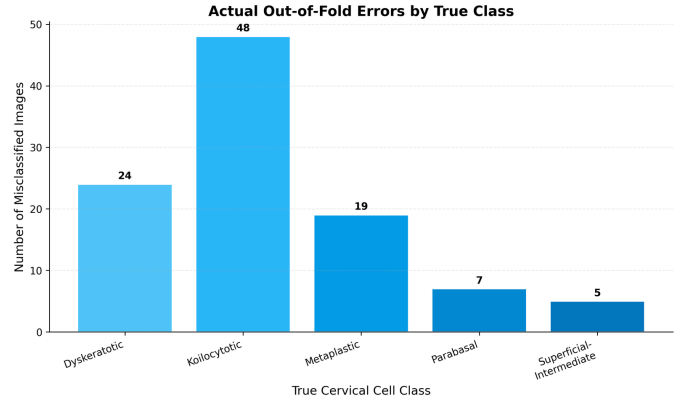
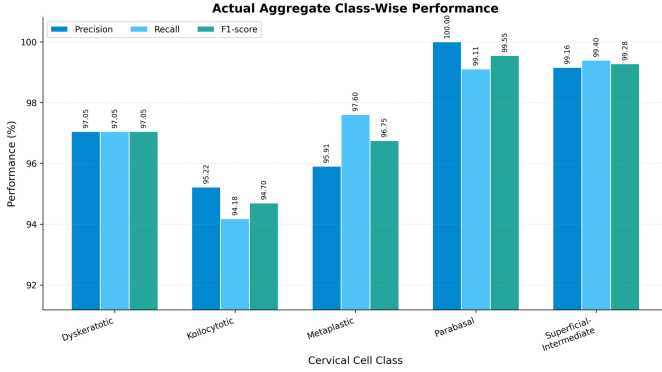
Table VII gives the pooled metrics for each class. Parabasal and Superficial-Intermediate cells show the most consistent recognition, while Koilocytotic cells obtain the lowest recall and F1-score and therefore account for the largest unresolved difficulty.

is

$$\frac{4.57 - 2.54}{4.57} \times 100 \approx 44.3\%. \quad (24)$$



Confusion-matrix counts. (a)
Row-normalized confusion matrix. (b)
Fig. 5. Aggregate out-of-fold confusion matrices, with true classes shown by rows and predicted classes by columns.



(a) Class-wise precision, recall, and F1. (b) Errors by true class.

Fig. 6. Class-level performance and error distribution.

TABLE VIII
MOST FREQUENT DIRECTED OUT-OF-FOLD ERRORS.

True → predicted	Images
Koilocytotic → Metaplastic	23
Dyskeratotic → Koilocytotic	20
Koilocytotic → Dyskeratotic	20
Metaplastic → Koilocytotic	14
Koilocytotic → Superficial-Intermediate	5

The pooled confusion matrix contains 48 errors for true Koilocytotic cells, 24 for Dyskeratotic, 19 for Metaplastic, 7 for Parabasal, and 5 for Superficial-Intermediate. Table VIII lists the most common error directions.

These results suggest that the hardest distinctions involve visually similar abnormal and benign morphologies, not the two normal classes that achieve the strongest performance.

Figure 7 makes the direction of each error explicit, unlike a condensed confusion summary. Koilocytotic examples are confused with both Metaplastic and Dyskeratotic cells, and examples from those classes are also predicted as Koilocytotic.

The pattern is bidirectional but not perfectly symmetric, which points to overlapping morphology in the learned feature space rather than a simple one-way label swap. A future experiment could treat these pairs as hard positive and hard negative examples in a supervised contrastive objective while retaining the same five-fold evaluation protocol.

E. Qualitative Grad-CAM Analysis

Grad-CAM is generated from the last EfficientNet-B0 feature block using gradients of the selected class logit. If A^q denotes channel q and o_c denotes the class logit, then

$$\beta_q^c = \frac{1}{HW} \sum_{i=1}^H \sum_{j=1}^W \frac{\partial o_c}{\partial A_{ij}^q}, \quad (25)$$

$$L_{\text{GradCAM}}^c = \text{ReLU} \left(\sum_q \beta_q^c A^q \right). \quad (26)$$

The gradient travels through the MLP, graph pooling, and both GAT layers before reaching the backbone. The resulting map

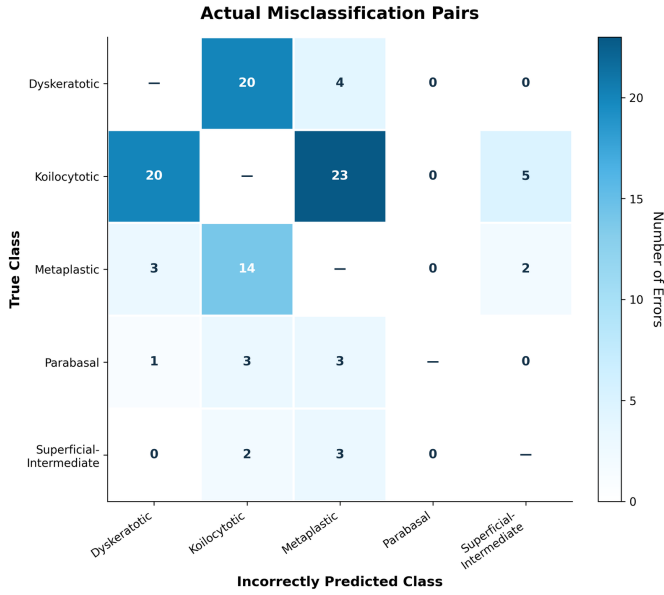


Fig. 7. Directed out-of-fold error pairs with diagonal entries removed. Each arrow points from the true class toward the predicted class.

therefore reflects the decision of the full model rather than that of a standalone CNN head.

For many correct predictions, the strongest activation lies around the nucleus or central cellular structure. Several errors instead highlight image borders, background regions, or broad cell outlines, indicating that the network may occasionally depend on non-specific cues. Grad-CAM should be viewed as an inspection tool rather than evidence of clinical reasoning, and a cytologist would need to judge whether the highlighted areas are medically relevant.

F. Difficult Correct Cases and High-Confidence Errors

For each class, the difficult-correct panel shows the held-out image that was classified correctly with the lowest normalized fuzzy confidence $s_{\hat{y}} / \sum_k s_k$. The corresponding values are 0.448, 0.432, 0.461, 0.450, and 0.542 for Dyskeratotic, Koilocytotic, Metaplastic, Parabasal, and Superficial-Intermediate cells. These cases show that challenging appearance—including weak contrast, blurred borders, small nuclei, and overlapping background content—does not necessarily result in an error.

The confident-error panel contains the ten misclassified samples with the highest normalized fuzzy confidence. Seven of these ten cases are true Koilocytotic cells. Across all 103 errors, 21 receive fuzzy confidence of at least 0.90 and 12 are above 0.95. Confidence by itself therefore cannot be used as proof that a prediction is correct.

G. Ablation and Component Analysis

A controlled ablation study was conducted to quantify the contribution of the main architectural components. Every trainable variant used the same SIPaKMeD dataset, stratified five-fold splits, random seed, augmentation, optimizer, learning rates, maximum epoch count, and early-stopping rule.

TABLE IX
CONTROLLED ABLATION RESULTS UNDER THE SAME FIVE-FOLD PROTOCOL.

Variant	Parameters	Acc. (%)	Std. (%)	Macro F1 (%)
Full model	4,286,243	97.4561	0.8924	97.4676
One GAT layer	4,260,003	97.4315	0.1639	97.4405
No GAT layers	4,233,763	97.4561	0.4032	97.4623
No global node	4,286,243	97.4314	0.4648	97.4403
No fuzzy reward	4,286,243	97.3573	0.5979	97.3699
EfficientNet-B0 only	4,013,953	97.6291	0.2391	97.6305

Consequently, only the component named by each variant was changed. Table IX reports the resulting parameter count, five-fold mean accuracy, fold-wise standard deviation, and macro F1-score.

The full model achieved 97.4561% mean accuracy and 97.4676% macro F1. Removing the second GAT layer reduced accuracy by only 0.0246 percentage points, while removing the global node reduced it by 0.0247 percentage points. Removing both GAT layers and directly averaging the five regional descriptors produced the same rounded mean accuracy as the full model, despite using 52,480 fewer trainable parameters. These results show that the GAT layers and global node did not provide a clearly measurable mean-accuracy advantage under the present protocol.

Removing the fuzzy reward from the training objective reduced mean accuracy to 97.3573% and increased the out-of-fold error count from 103 to 107. This 0.0988-percentage-point reduction was the largest change among the internal component-removal experiments, indicating a small positive training effect from the fuzzy reward. Nevertheless, its magnitude remained modest relative to the observed fold-wise variability.

The pure EfficientNet-B0 baseline removed the 1×1 projection, region graph, global graph node, both GAT layers, fuzzy head, and fuzzy reward. It achieved the highest mean accuracy of 97.6291%, a macro F1-score of 97.6305%, and 96 out-of-fold errors, while using 272,290 fewer parameters than the full model. Therefore, the ImageNet-pretrained EfficientNet-B0 backbone was the dominant source of predictive performance, and the additional graph-fuzzy components did not improve mean accuracy over the simpler backbone in these experiments.

The post-training fuzzy decision rule was also examined analytically. For class k , the implemented score is

$$s_k = (\gamma_c p_k) [\gamma_s \text{clip}(p_k + m, 0, 1)], \quad (27)$$

where m is the same top-two probability margin for every class in a sample. When $\gamma_c \gamma_s > 0$, this expression is monotonic in p_k , and hence

$$\arg \max_k s_k = \arg \max_k p_k. \quad (28)$$

A numerical verification over one million randomly generated five-class probability vectors produced 100% agreement and zero disagreements between softmax argmax and fuzzy-score

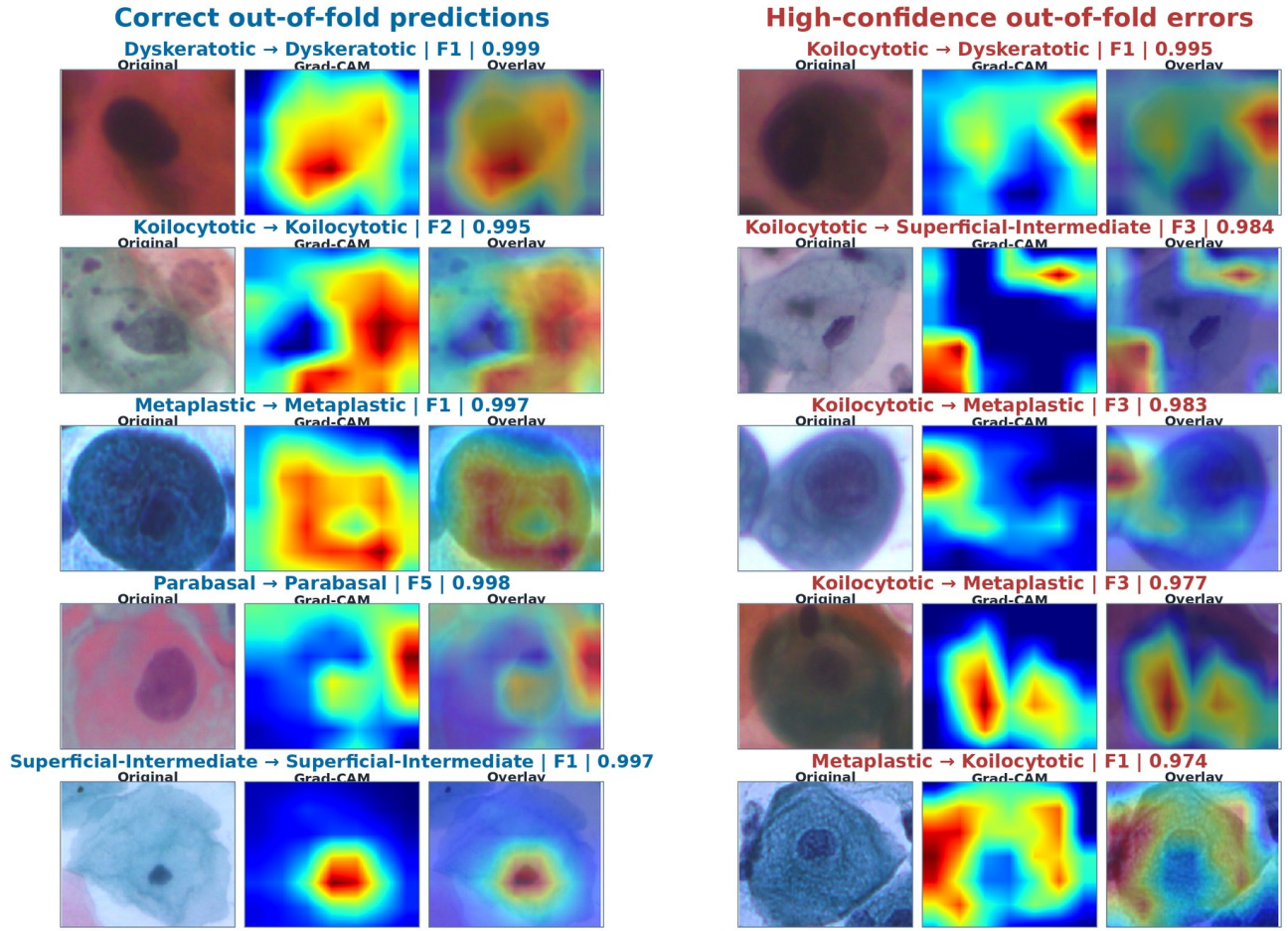


Fig. 8. Out-of-fold Grad-CAM visualizations. One correctly classified example from each class appears on the left, while the five most confident errors appear on the right.

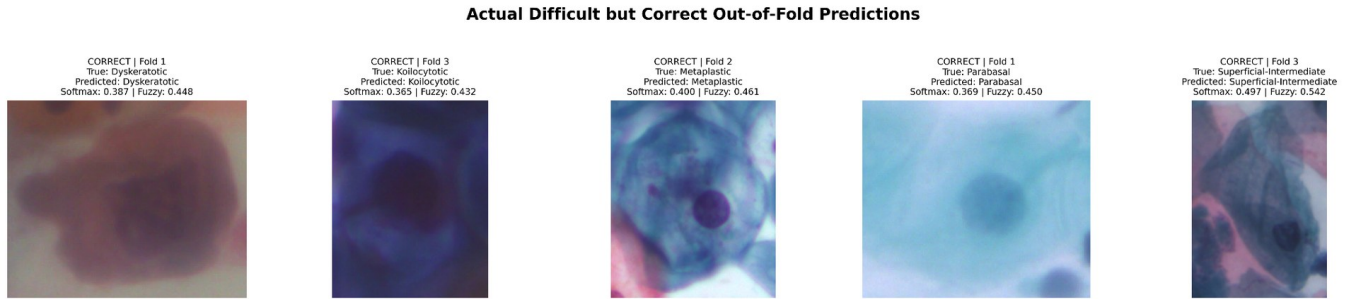


Fig. 9. Difficult yet correct out-of-fold cases, chosen as the lowest-confidence correct prediction within each true class.

argmax. Thus, the fuzzy transformation does not change the final class ordering under positive shared scales; its observable contribution arises from the fuzzy reward used during training.

Overall, all variants retained strong classification performance, but the differences between the graph-based variants were smaller than their fold-wise variation. The component effects should therefore be interpreted as marginal empirical trends rather than statistically established improvements.

H. Why the Proposed Model Performs Well

Four aspects of the design help explain the measured performance. First, end-to-end fine-tuning adjusts EfficientNet features to the visual characteristics of cervical cells. Second, the region graph preserves four local descriptors instead of collapsing the spatial map immediately. Third, graph attention combines local information with global context through weights that vary from one image to another. Finally, augmentation, separate learning rates, label smoothing, dropout, AdamW, early stopping, and mixed-precision training provide

Actual Out-of-Fold Misclassifications

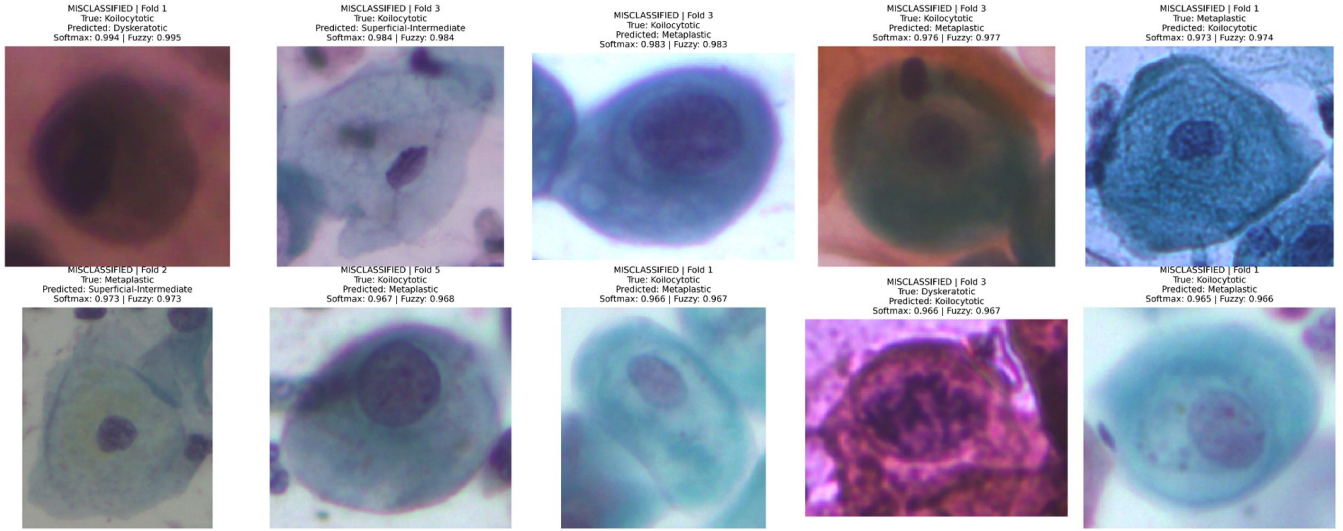


Fig. 10. The ten most confident out-of-fold errors. These examples are intended to reveal overconfident failure modes rather than typical mistakes.

a stable optimization procedure.

I. Limitations and Threats to Validity

The results should be interpreted in light of the following limitations:

- 1) The baseline comparison uses values from a separate study rather than a paired rerun on the same folds.
- 2) The held-out fold is used both to select the checkpoint and to report final performance; nested validation would lessen this source of selection bias.
- 3) The split does not enforce grouping by patient, slide, or acquisition source.
- 4) Every image is processed with the same fixed graph structure.
- 5) Dividing a 7×7 feature map produces quadrants of unequal size.
- 6) The implemented GAT uses a single attention head and the original static GAT formulation.
- 7) The fuzzy head contains two shared scalars rather than class-specific membership functions.
- 8) The confidence scores are not fully calibrated, and several wrong predictions remain highly confident.
- 9) Grad-CAM has limited spatial resolution and cannot demonstrate clinical causality.
- 10) The study is restricted to one isolated-cell dataset and does not include external laboratories, whole-slide images, or prospective clinical evaluation.

VII. CONCLUSION AND FUTURE WORK

This study introduced an EfficientNet-B0 Region-Graph Attention Network with a Fuzzy Confidence Head for five-class cervical cytology classification, following the supplied implementation closely. Instead of using a fuzzy ensemble with three separate backbones, the model relies on one EfficientNet-B0 encoder, reduces the final feature map to 160

channels, forms four local nodes and one global node, applies two graph attention updates, and produces its decision through graph pooling, an MLP, and a fuzzy confidence-and-separation score.

Across all 4,049 SIPaKMeD images, the stratified five-fold experiment reached mean accuracy of 97.46%, macro precision of 97.48%, macro recall of 97.47%, and macro F1-score of 97.47%. The combined out-of-fold predictions include 3,946 correct classifications and 103 errors. Compared descriptively with the published five-class result of the assigned paper, the proposed mean is higher by 2.03 percentage points and corresponds to an approximate 44.3% reduction in error rate. Parabasal and Superficial-Intermediate cells are recognized most reliably, whereas Koilocytotic cells account for the largest share of the remaining confusion. The qualitative results show frequent attention to central cellular structures but also reveal background-sensitive decisions and confidently wrong predictions.

Future experiments should reproduce the assigned baseline on the same folds, complete the controlled retraining studies listed in Table IX, and use nested cross-validation that respects patient or slide groups. Other useful directions include learning the graph connectivity from each image, comparing the current GAT with GATv2, multi-head, and residual alternatives, developing class-specific fuzzy memberships, calibrating confidence, measuring latency, memory use, and energy consumption, and testing on Mendeley LBC, Herlev, whole-slide images, and data from external laboratories. At this stage, the model is best regarded as an experimental decision-support approach rather than a clinically validated diagnostic tool.

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